Data Analytics

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Agenda

Must include the following, but not limited to:

1. About the company
2. Assigned Project details
3. Technologies used
4. Team and role
5. Results and snapshots
6. Conclusion
7. Internship Completion certificate/status

1. About the Company

- **CodTech** is a platform that provides industry-oriented internship programs in various technical domains such as Data Analysis, Machine Learning, and Web Development.

- The organization focuses on practical learning by assigning real-world tasks that help students gain hands-on experience.

- The internship program is designed to enhance technical skills, analytical thinking, and problem-solving abilities of students.

2. Assigned Project

The internship project includes multiple data analysis tasks:

- Exploratory Data Analysis (EDA)
- House Price Prediction (Machine Learning)
- Sales Dashboard Development
- Sentiment Analysis using NLP

Goal:
To understand real-world data handling and build practical solutions

3. Technologies Used

Programming Language:

- Python

Libraries:

- Pandas – data manipulation
- NumPy – numerical operations
- Matplotlib & Seaborn – visualization
- Scikit-learn – machine learning

Tools:

- VS Code
- Jupyter Notebook
- Dash & Plotly (Dashboard)

RESULT
Task1: Big Data Analysis

Description:
Performed exploratory data analysis on dataset

Steps Performed:

- Loaded dataset using Pandas
- Checked data using .info() and .describe()
- Handled missing values
- Created visualizations

Result:
Identified patterns and trends

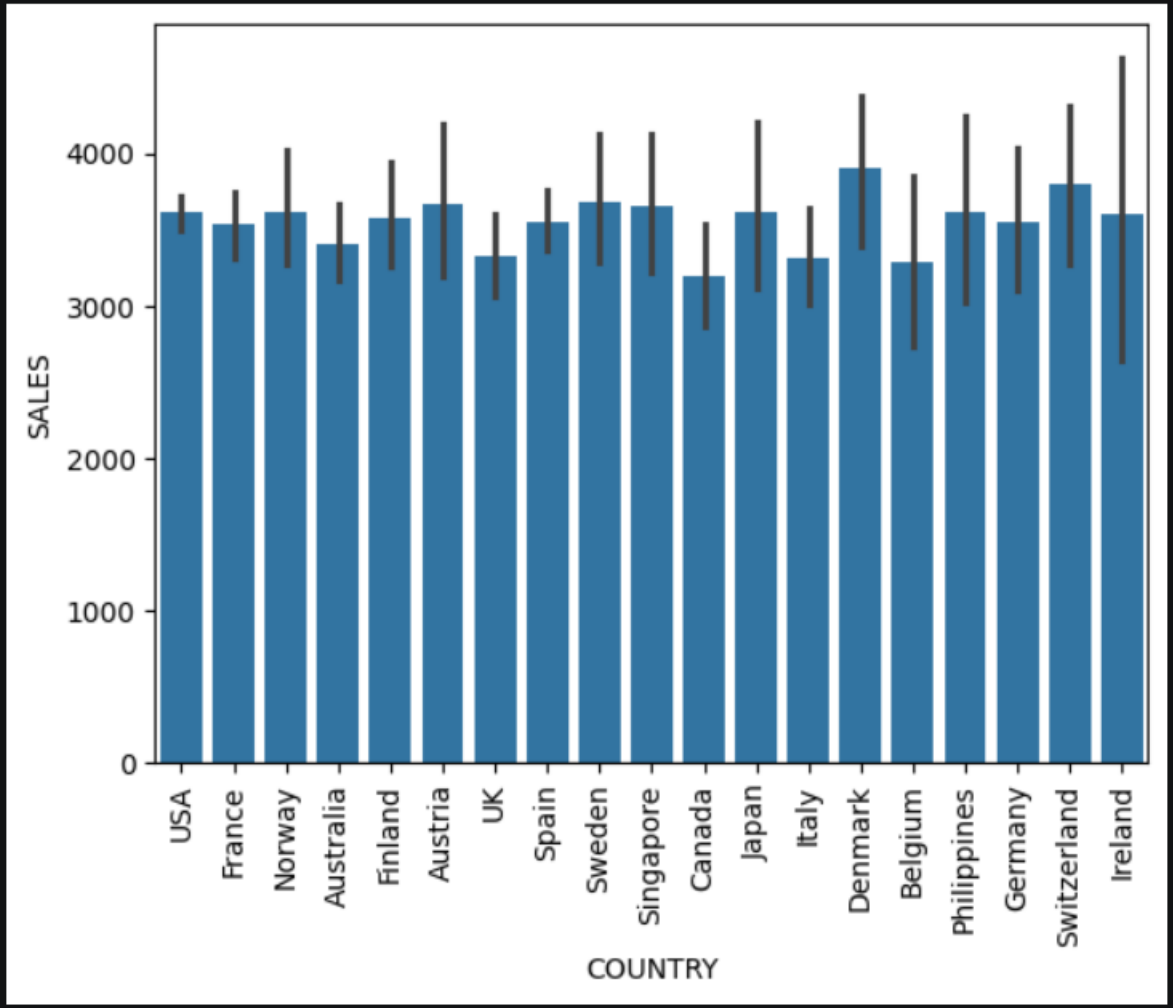
Outcome:
Learned data preprocessing and visualization

```python
import seaborn as sns
import matplotlib.pyplot as plt

sns.barplot(x="COUNTRY", y="SALES", data=pdf)
plt.xticks(rotation=90)
plt.show()
```

[10]

...



Task 2: Machine Learning(House Price Prediction)

Description:
Built ML model to predict house prices

Steps Performed:

- Selected features and target variable
- Split dataset (train-test)
- Applied Linear Regression
- Evaluated using MSE & R^2 score

Result:
Model achieved good prediction accuracy

Outcome:
Learned model training and evaluation

```python
mse = mean_squared_error(y_test, predictions)
r2 = r2_score(y_test, predictions)

print("Mean Squared Error:", mse)
print("R2 Score:", r2)
```

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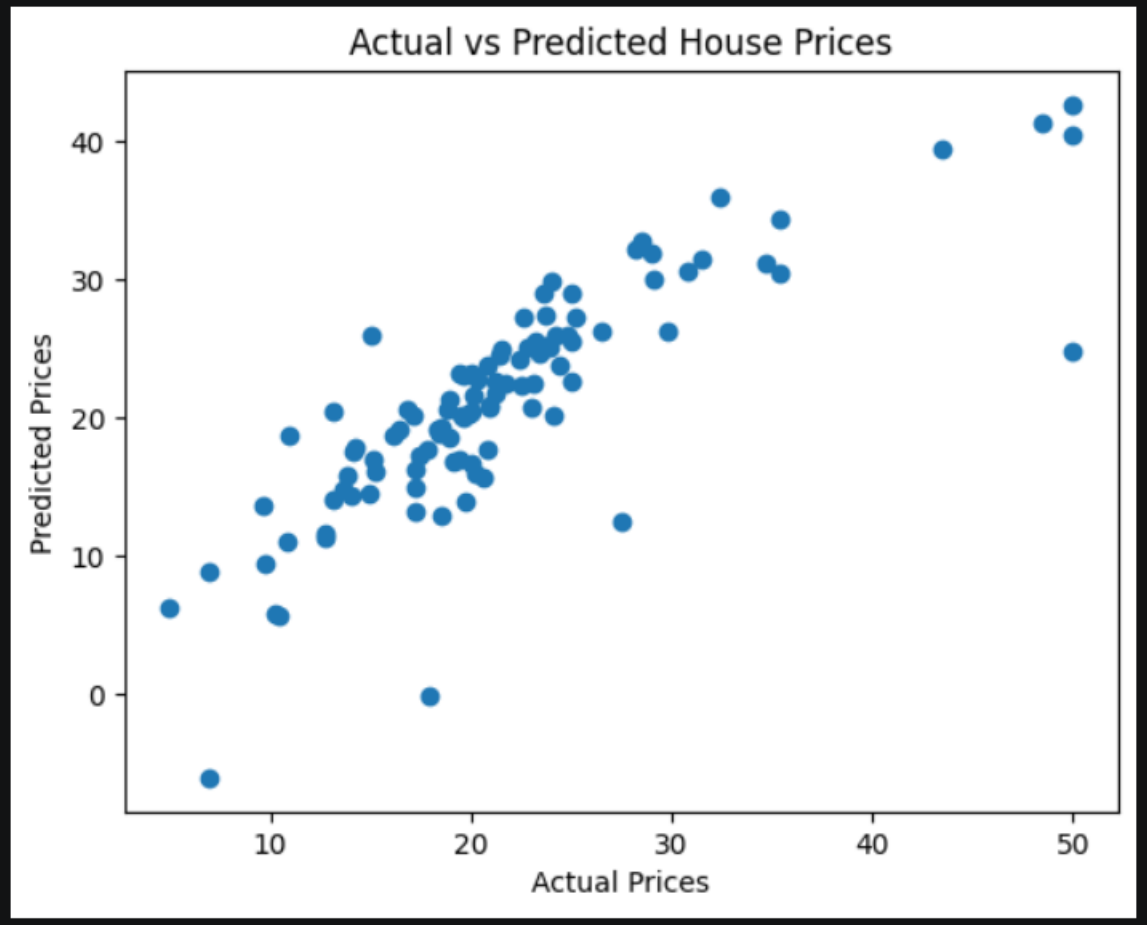
```
Mean Squared Error: 24.291119474973538
R2 Score: 0.6687594935356317
```

```python
plt.scatter(y_test, predictions)
plt.xlabel("Actual Prices")
plt.ylabel("Predicted Prices")
plt.title("Actual vs Predicted House Prices")
plt.show()
```

[10]



Task 3: Dashboard Development

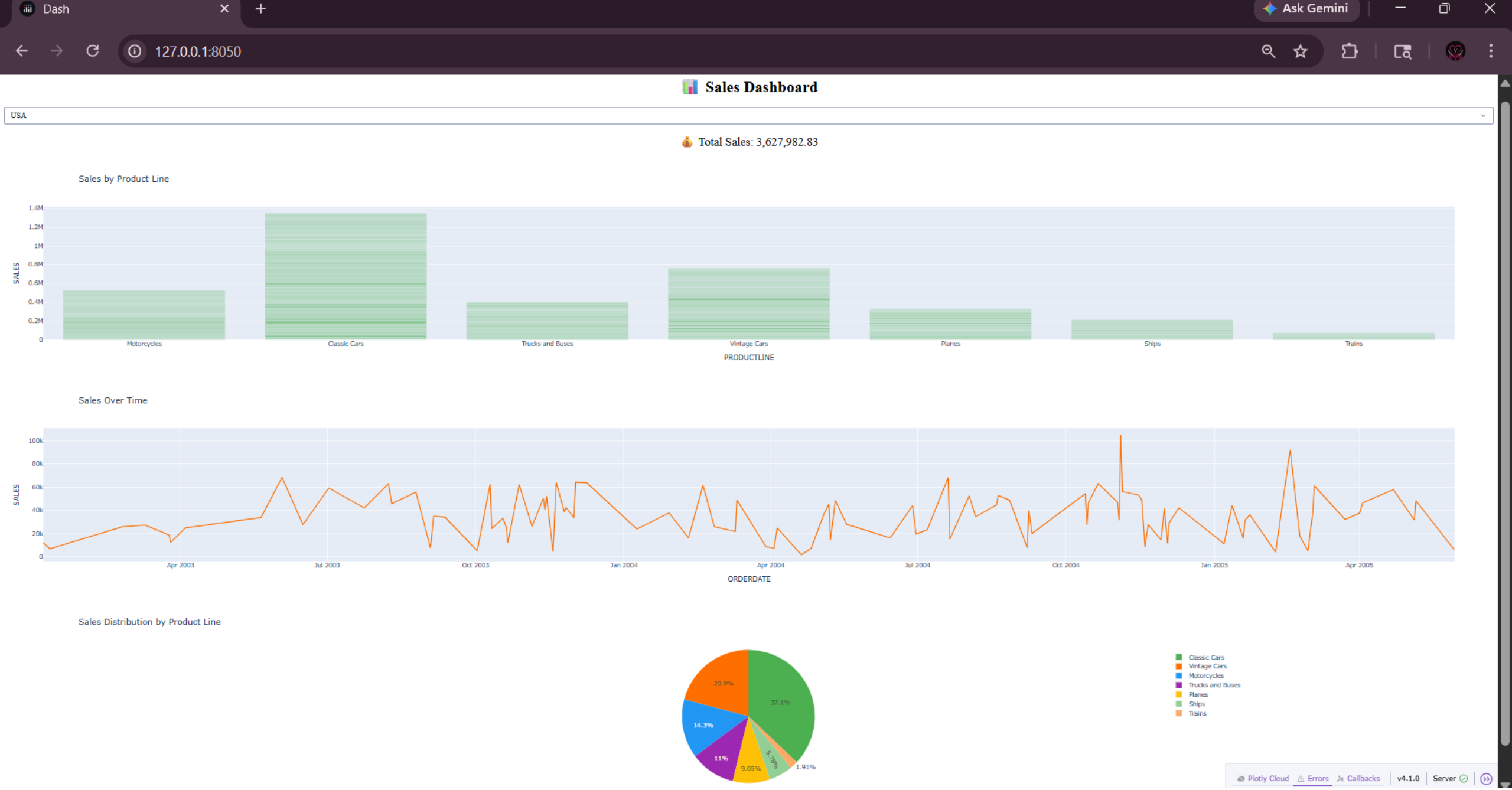
Description:
Developed interactive sales dashboard

Steps Performed:

- Loaded dataset
- Created bar, line, and pie charts
- Used Dash & Plotly
- Added dropdown filters

Result:
Interactive dashboard showing insights

Outcome:
Improved data visualization skills


Dash
Ask Gemini
127.0.0.1:8050
Sales Dashboard
USA
Total Sales: 3,627,982.83
Sales by Product Line
SALES
1.4M
1.2M
1M
0.8M
0.6M
0.4M
0.2M
0
Motorcycles
Classic Cars
Trucks and Buses
Vintage Cars
Planes
Ships
Trains
PRODUCTLINE
Sales Over Time
SALES
100k
80k
60k
40k
20k
0
Apr 2003
Jul 2003
Oct 2003
Jan 2004
Apr 2004
Jul 2004
Oct 2004
Jan 2005
Apr 2005
ORDERDATE
Sales Distribution by Product Line
20.9%
37.1%
14.3%
11%
9.05%
5.78%
1.91%
Classic Cars
Vintage Cars
Motorcycles
Trucks and Buses
Planes
Ships
Trains
Plotly Cloud
Errors
Callbacks
v4.1.0
Server

Task 4: Sentiment Analysis

Description:
Classified reviews into positive & negative

Steps Performed:

- Cleaned text data
- Applied TF-IDF Vectorization
- Used Logistic Regression
- Evaluated accuracy (~91%)

Result:
Model successfully classified sentiments

Outcome:
Learned NLP basics and text classification

```python
#MODEL TRAINING
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=1000)
model.fit(X_train_vec, y_train)

#PREDICTION
y_pred = model.predict(X_test_vec)

#EVALUATION
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

[ ]

```
Accuracy: 0.91424
```

Generate

```python
review = ["This movie was absolutely amazing and I loved it"]

# transform using same vectorizer
review_vec = vectorizer.transform(review)

# predict
prediction = model.predict(review_vec)

if prediction[0] == 1:
    print("Positive Review 😊")
else:
    print("Negative Review 😞")
```

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```
Positive Review 😊
```

```python
review = ["worst movie ever"]

# transform using same vectorizer
review_vec = vectorizer.transform(review)

# predict
prediction = model.predict(review_vec)

if prediction[0] == 1:
    print("Positive Review 😊")
else:
    print("Negative Review 😞")
```

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```
Negative Review 😞
```

4. Conclusion

- Completed full data analysis pipeline
- Worked on real-world datasets
- Built ML models and dashboards
- Performed sentiment analysis

Skills gained:

- Python
- Machine Learning
- Data Visualization
- NLP

5. Internship Completion certificate/status

- All 4 tasks completed successfully
- Project submitted
- Certificate: Awaited

Thank you